Personal Portfolio Website (`rah3.github.io`)

Rahie Datta

Carleton University - Bachelor of Computer Science

Live Site: https://rah3.github.io
GitHub Repository: https://github.com/rah3/rah3.github.io

1 About

I am a fourth-year Computer Science student at **Carleton University** pursuing a Bachelor of Computer Science with a Minor in Mathematics.

Core technical interests and focus areas include:

- Software Engineering & System Design
- Linux Systems & Process Management
- Backend Development & APIs
- Database Design & Optimization
- Telecommunications Systems
- Automation & Scripting

2 Technology Stack

The project relies on a modern, lightweight, and framework-free JavaScript architecture. It uses Node.js and Express solely for local development ergonomics, while the production assets are fully optimized for static delivery via GitHub Pages.

2.1 Frontend

- **HTML5 / CSS3**: Semantic markup and modern custom styling.
- **Vanilla JavaScript (ES6+)**: Modular, performant implementation without heavy framework overhead.

2.2 Backend & Tooling

- **Node.js & Express**: Local development server and static asset proxying.
- **Git & GitHub**: Version control and CI/CD hosting via GitHub Pages.
- **Nodemon**: Hot-reloading utility during local development.

3 Architectural Style

The codebase implements a **modular component pattern** and functional composition. Rather than relying on heavy client-side frameworks, the UI components are isolated into independent, testable modules.

3.1 Core Modules

- `Navigation.js`: Manages mobile responsive toggle states and layout adjustments.
- `ScrollSpy.js`: Tracks user scroll position to dynamically highlight active navigation links.
- `EmailClipboard.js`: Handles user interactions for quick email copying with visual feedback.

3.2 Application Factory Pattern

Application startup and lifecycle management are initialized through a lightweight factory pattern:

```
createApp().init();
```

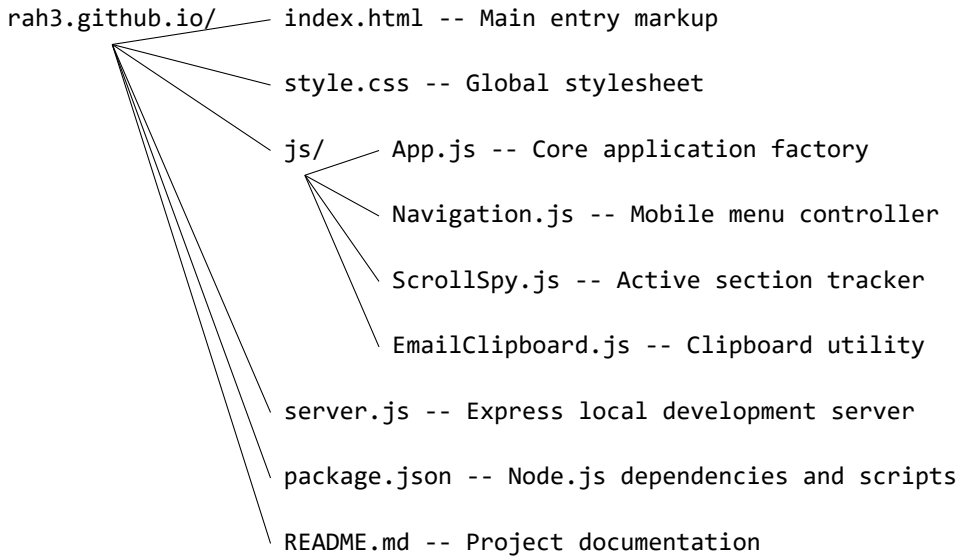
4 Features

Here are the features that make this portfolio website stand out:

- **Fully Responsive**: Mobile-first design adapting seamlessly across mobile, tablet, and desktop viewports.
- **Active Section Highlighting**: Real-time viewport tracking using intersection observers / scroll spy.
- **Interactive Utilities**: One-click email copy-to-clipboard functionality.
- **Lightweight & Fast**: Zero frontend framework bloat ensuring instant page load speeds.
- **Express Development Server**: Local mock environment mirroring production static behavior.

5 Project Structure

Below is a visual representation of the repository structure:



6 Featured Projects

Below are some of the notable projects I have worked on during my academic career:

1. Telecommunications Switching Database (COMP 3005)
2. Linux Systems & Shell Development (COMP 3000)
3. Professional Portfolio Website (rah3.github.io)

6.1 Telecommunications Switching Database

- **Course:** COMP 3005 (Database Management Systems)
- **Technologies:** SQLite, SQL, Relational Database Design
- **Overview:** Designed and implemented a robust relational database schema tailored for telecommunications routing and subscriber switching management.

6.2 Linux Systems & Shell Development

- **Course:** COMP 3000 (Operating Systems)
- **Technologies:** Linux, Bash, System Calls, C
- **Overview:** Explored low-level OS primitives, process management (`fork`, `execve`), environment variable handling, and remote SSH administration.

6.3 Professional Portfolio Website

- **Technologies:** Node.js, Express, JavaScript, HTML5/CSS3
- **Overview:** This repository was built from scratch to highlight academic and practical engineering work.

7 Local Development

To run the development environment locally:

1. **Clone the repository:**

```
git clone https://github.com/rah3/rah3.github.io.git
cd rah3.github.io
```

2. **Install dependencies:**

```
npm install
```

3. **Start the development server:**

```
npm run dev
```

   (Or standard start: `npm start`)

4. **Open in your browser:**
   Navigate to http://localhost:3000.

8 Deployment

This repository uses a decoupled deployment model:

- **Production:** The contents of the `public/` directory are deployed statically to **GitHub Pages**.
- **Development:** The `server.js` Express backend is used exclusively for local testing and does not run in production.

9 Author

Rahie Datta
Carleton University - Bachelor of Computer Science (BCS)
Expected Graduation: 2027
GitHub: https://github.com/rah3